

Evaluating Tokenization Mismatch in Slovak Morphology

Adam Gábor
FIIT STU

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Abstract

Subword tokenizers, the primary interface for modern language models, are commonly evaluated using compression-oriented metrics like token count or fertility. In morphologically rich fusional languages such as Slovak, high compression may merge linguistic units arbitrarily, breaking the boundaries between morphemes and inflectional suffixes. In this paper, we analyze multilingual and language-specific subword tokenizers using a multi-layered framework that combines lemma-level morphemic segmentation, stem-suffix alignment over large-scale heuristically generated inflected forms, and a newly curated open tokenization evaluation benchmark. Our findings reveal a systematic trade-off between sequence compression and morphological alignment: highly compact tokenizers often achieve efficiency by disrupting functional boundaries, whereas general-purpose tokenizers preserve morphological units more reliably at the cost of longer token sequences. Because morphological boundaries encode grammatical information, tokenization mismatch may affect how language models process and generate inflected words. Our work provides practical criteria for tokenizer selection and vocabulary design in Slavic NLP and establishes a reproducible foundation for developing more linguistically robust language models.

1 Introduction

1.1 Problem Statement

Modern transformer-based language models operate on tokenized text rather than directly on raw character or word sequences [1, 2]. In contemporary NLP, this interface is usually provided by subword tokenizers such as Byte-Pair Encoding (BPE), WordPiece, or Unigram/SentencePiece, which segment text into discrete vocabulary units while reducing the out-of-vocabulary problem [3, 4, 5]. These methods are effective engineering solutions: they allow neural models to represent rare and unseen words through smaller recurring units, while keeping the vocabulary finite.

However, subword vocabularies are usually learned from corpus statistics rather than from explicit linguistic structure. As a result, the boundaries introduced by a tokenizer do not necessarily align with morpheme boundaries, stems, or inflectional suffixes. Recent work has argued that tokenizer quality should not be evaluated only through compression-oriented measures, but also through linguistic alignment with morphological structure [6, 7, 8]. This issue is especially important for morphologically rich, fusional languages, where

grammatical information is encoded inside word forms. A tokenizer may produce short token sequences while still merging together stems and functional endings, or it may split words into more tokens while preserving linguistically meaningful boundaries.

Traditional tokenizer evaluations often emphasize efficiency metrics such as token count, average sequence length, or fertility. These metrics are important because tokenization directly affects computational cost, inference speed, context usage, and model input length. However, they do not tell us whether the resulting token boundaries preserve the morphological units that are relevant for grammatical interpretation. For Slovak, this distinction is central: a compact tokenizer is not necessarily a morphologically aligned tokenizer.

The practical concern is compositional generalization. If related Slovak forms share visible stems or suffixes, a model can in principle reuse evidence across a paradigm, for example by associating endings such as *-ami* with recurring inflectional functions. If the same ending is repeatedly absorbed into different larger tokens, this regularity becomes less explicit in the token sequence. We therefore treat boundary preservation as a property that may support downstream morphological processing, while leaving its direct effect on model accuracy for future work.

This leads to the main research question of the paper: do tokenizers that are efficient for Slovak noun forms also preserve morphological boundaries, or do compression and boundary alignment behave as separate evaluation axes?

1.2 Why Slovak Morphology Matters

Slovak, as a Slavic language, has a rich fusional morphology. Nouns inflect for six productive cases: nominative, genitive, dative, accusative, locative, and instrumental. These forms are distributed across singular and plural number and across masculine, feminine, and neuter gender classes. The language follows a diverse set of declension patterns (*vzory*) documented in Slovak morphological literature [9, 10]. This makes Slovak noun declension a useful setting for studying whether subword tokenizers preserve grammatically meaningful boundaries.

In Slovak, the boundary between the stem and the inflectional suffix is often informative because even short endings such as *-o*, *-i*, *-a*, or *-e* may encode case, number, gender, and paradigm information. If a tokenizer merges such endings with stem material, the grammatical boundary becomes hidden at the token level. A less compact tokenization may still be preferable for morphological analysis if it preserves the stem-suffix boundary.

For example, the inflected word *stromami* can be analyzed as stem *strom* plus the instrumental plural suffix *-ami*. The tokenization [“strom”, “ami”] exposes this boundary, whereas [“stro”, “mami”] crosses it and [“stromami”] keeps it hidden inside one token. We therefore use Slovak noun morphology to evaluate whether tokenizers preserve inflectional structure rather than only compressing words efficiently.

1.3 Contributions

In this work, we systematically investigate the alignment between subword tokenization and morphological boundaries in Slovak. We develop a multi-layered evaluation framework that separates efficiency from morphological fidelity. At the lemma level, we evaluate tokenizer behavior against morphemic boundaries derived from the Slovak Root Morpheme Dictionary (SKMS) [11]. At the inflected-form level, we evaluate whether tokenizers pre-

serve practical stem–suffix boundaries across 332,931 heuristically generated Slovak noun forms.

We apply this framework to twelve tokenizers, including OpenAI tokenizers, multilingual LLM tokenizers, multilingual encoder tokenizers, and Slovak-specific systems. This comparison includes SlovakBERT [12] and the Slovak Morphological Tokenizer (SKMT) [13], allowing us to compare general-purpose multilingual tokenizers with tokenizers designed specifically for Slovak language processing.

Our results show that tokenization efficiency and morphological boundary preservation are two distinct dimensions of tokenizer quality. This supports the need for boundary-aware tokenizer evaluation in morphologically rich languages.

We introduce SlovakTokenBench, a curated 1,000-item benchmark for reproducible evaluation of Slovak noun tokenization, accompanied by 50 separate simple examples for quick inspection. The benchmark combines general examples, paradigm-representative forms, hard tokenization cases, and targeted morphological edge cases. It is intended as a compact evaluation resource for future work on Slovak and Slavic tokenizer design.

2 Background and Motivation

2.1 Subword Tokenization

We use subword tokenization as an umbrella term for methods that segment text into units smaller than full words but larger than individual characters. BPE builds a vocabulary by iteratively merging frequent symbol pairs, WordPiece uses a related vocabulary-learning strategy popularized by BERT-style models, and Unigram/SentencePiece selects segmentations from a probabilistic vocabulary [3, 4, 5, 2]. These methods produce the token sequences consumed by language models, but their units should not be assumed to correspond to linguistic morphemes [14].

2.2 Why Token Count Is Not Enough

Many tokenizer comparisons focus on efficiency: how many tokens are needed to encode a word, a sentence, or a whole dataset. These metrics matter because longer token sequences increase computational cost, reduce effective context length, and may create unequal costs across languages [15]. Fertility, usually understood as the number of tokens per word or relative to word length, is therefore a useful measure of tokenization efficiency.

Efficiency alone, however, does not tell us whether the produced tokens are linguistically useful. A tokenizer may encode a Slovak word using only one or two tokens, while still hiding the boundary between the lexical stem and the grammatical suffix. Another tokenizer may produce a longer sequence while preserving an important morpheme boundary. For morphologically rich languages, this means that token count and morphological alignment must be treated as separate dimensions of tokenizer quality [6, 8].

2.3 Slovak Morphology

Slovak is a useful test case because it combines rich inflection with limited tokenizer-specific evaluation resources. Its noun system is well described in Slovak morphological literature, including detailed accounts of declension patterns and inflectional forms [9, 10]. At the same time, the Slovak Root Morpheme Dictionary (SKMS) provides morphemic

segmentation for Slovak word structure, giving us a reference point for comparing tokenizer boundaries with linguistically correct boundaries [11].

This combination makes Slovak practical for boundary-aware evaluation: the language has complex enough morphology for tokenization mismatches to matter, but also has linguistic resources to define explicit reference boundaries. A tokenizer that merges Slovak endings with stem material may still be efficient, but it hides grammatical structure that is relevant for inflectional analysis.

2.4 Efficient vs. Morphologically Aligned Tokenization

The tension between compression and alignment has motivated recent work on morphology-aware and lexically grounded tokenization. MorphBPE modifies BPE with explicit morphological information, while lexically grounded subword segmentation evaluates tokenization partly through morpheme-boundary precision [7, 16]. For Slovak, SKMT is especially relevant because it incorporates Slovak morphological information into BPE-based tokenizer training and reports strong root-integrity results [13]. Rather than treating efficiency and alignment as mutually exclusive, our evaluation treats them as two distinct dimensions of tokenizer performance.

2.5 Motivation for Evaluating Slovak Tokenizers

Existing Slovak language model work already shows the need for Slovak-specific evaluation resources [12]. Tokenizer evaluation for Slovak morphology remains less developed than general model evaluation, which motivates an intrinsic tokenizer study before downstream model testing.

3 Data

3.1 Source Lexicon

Our source data comes from a cleaned Slovak noun lexicon derived from SKMS morphemic annotations. The table provided by the Ľ. Štúr Institute of Linguistics of the Slovak Academy of Sciences contains 30,913 noun entries with lemmas, morphemic segmentations, gender, declension pattern, and cleaning flags. For the lemma-level experiment, we use the parsed morpheme boundaries from this table as reference boundaries for tokenizer evaluation [11].

3.2 Generated Inflected Forms

To evaluate tokenization on actual declined forms, we generated Slovak noun forms across case and number using declension-pattern information and heuristic rules for common Slovak alternations. These heuristics cover regular paradigm endings and frequent stem changes such as mobile vowels, selected consonant alternations, and foreign suffix patterns. The full generated table contains 370,920 candidate forms. After filtering for the main evaluation, we use 332,931 forms, covering 200,278 unique surface forms with 25,906 lemmas. Each evaluated form includes a stem, suffix, and stem-suffix boundary. These generated forms are used as a large-scale evaluation resource, not as a fully manually validated morphological dictionary. We therefore interpret aggregate results as evidence

about tokenizer behavior over a generated evaluation set, and rely on the checked subset for the reusable release benchmark.

3.3 Benchmark Subset

We also construct SlovakTokenBench, a fixed 1,000-item benchmark for reproducible tokenizer comparison, with 50 separate simple examples for quick tests and inspection. It contains Slovak noun forms with grammatical metadata and stem–suffix boundary annotations. The original surface forms were manually checked. The replacement forms were selected from paradigm entries marked as accepted and passed the same reconstruction and boundary checks.

4 Metrics

4.1 Token Count and Fertility

We use token count as the simplest efficiency metric: it is the number of tokens produced for a given word form. Lower token count means that the tokenizer represents the form more compactly and efficiently [15].

Because word forms differ in length, we also report a character-normalized fertility score. In much of the literature, fertility is reported as tokens per word; in this paper we use token count divided by character count so that short and long Slovak forms can be compared more directly:

$$\text{fertility}(w, T) = \frac{\text{token_count}(w, T)}{\text{character_count}(w)}.$$

4.2 Morpheme Boundary Crossing Rate

Our main morphology-aware metric is Morpheme Boundary Crossing Rate (MBCR). Given a set of reference boundaries, MBCR measures the proportion of boundaries that fall inside tokenizer spans:

$$\text{MBCR}(w, T) = \frac{\text{crossed_gold_boundaries}(w, T)}{\text{total_gold_boundaries}(w)}.$$

A boundary is crossed when a token starts before the boundary and ends after it. Lower MBCR is better: a score of 0 means that no evaluated boundary was crossed, while a score of 1 means that every evaluated boundary was crossed.

For lemma-level evaluation, the reference boundaries come from SKMS morphemic segmentation [11]. For inflected-form evaluation, the reference boundary is the generated stem–suffix boundary. We also report morphological alignment as the complementary score $1 - \text{MBCR}$, where higher values indicate better boundary preservation. Similar boundary-aware evaluation has been used in recent work on morphological tokenizer alignment [6, 16].

4.3 Stem–Suffix Preservation

For inflected forms, we additionally evaluate whether the tokenizer preserves the practical stem–suffix boundary. The strict stem-boundary match is a binary metric: it is 1 if the

tokenizer places a token boundary exactly at the reference stem–suffix boundary, and 0 otherwise.

We also report a graded stem preservation score. It is 1 when all reference stem–suffix boundaries are preserved, 0 when at least one reference boundary is crossed by a token, and 0.5 only for an offset-alignment anomaly where the boundary is neither recovered nor reliably enclosed by a token span. The corrected evaluator uses explicit character-safe offset handling and does not silently assign this fallback to an ordinary no-prediction case.

Let G be the set of reference stem–suffix boundaries, P the set of predicted token boundaries, and C the set of reference boundaries crossed by tokenizer spans. The strict score is:

$$\text{Strict}(w, T) = \begin{cases} 1 & \text{if } G \subseteq P, \\ 0 & \text{otherwise.} \end{cases}$$

The graded score is:

$$\text{Graded}(w, T) = \begin{cases} 1.0 & \text{if } G \subseteq P, \\ 0.0 & \text{if } C \neq \emptyset, \\ 0.5 & \text{otherwise.} \end{cases}$$

The third state is reserved for incomplete or irregular offset alignment. In the released benchmark, rows with a non-empty gold boundary and no predicted boundary receive zero for precision, recall, and F1; boundary metrics are undefined only when the gold set is empty.

4.4 Boundary Precision, Recall, and F1

We compute boundary precision, recall, and F1 by comparing predicted token boundaries with reference linguistic boundaries. Precision measures how many predicted token boundaries are reference boundaries, while recall measures how many reference boundaries are recovered by token boundaries. F1 summarizes the two.

A tokenizer may introduce many extra token boundaries and still avoid crossing the main stem–suffix boundary, or it may be compact while crossing a linguistically important boundary. We therefore interpret the results using two separate axes: efficiency, measured by token count and fertility, and morphological fidelity, measured by MBCR, strict stem–boundary match, graded stem preservation, and boundary F1.

5 Experimental Setup

5.1 Tokenizer Selection and Categories

We evaluate twelve tokenizers chosen to cover several practical tokenizer families used in current NLP and LLM systems. The set includes OpenAI tokenizer vocabularies (cl100k_base, o200k_base), Slovak-specific or Slovak-relevant tokenizers (SlovakBERT and SKMT), multilingual encoder tokenizers (mBERT, XLM-RoBERTa, mDeBERTa), and modern multilingual LLM tokenizers (Qwen3, Mistral NeMo Tekken, Gemma 3, Llama 3.1, and Llama 4).

The selected tokenizers also differ in vocabulary design and effective granularity. Vocabulary size, training data, and language specialization jointly influence whether Slovak

forms are represented as whole words, large subwords, or many smaller fragments. Larger or more Slovak-adapted vocabularies can reduce fertility by storing frequent Slovak units, but this may also hide internal morphological boundaries. Conversely, tokenizers with less suitable Slovak coverage may split words more aggressively, sometimes improving boundary scores through over-segmentation rather than true morphological alignment.

This selection is intended to compare compact general-purpose tokenizers with tokenizers trained for multilingual or Slovak-specific settings. SlovakBERT is included as the most prominent Slovak language-model baseline [12], while SKMT is included because it explicitly incorporates Slovak morphological information into tokenizer training [13].

5.2 Tokenization-Only Evaluation and Offset Alignment

The evaluation is tokenization-only: we do not run the full language models and do not measure downstream task accuracy. Each tokenizer is applied to the same Slovak noun forms, and the resulting token boundaries are compared with reference morphemic or stem-suffix boundaries. This lets us isolate tokenizer behavior from model architecture, training data, and decoding strategy.

For each tokenized form, we convert tokenizer output into character offsets. When a tokenizer provides reliable offset mappings, we use them directly. When offsets are not exposed in a compatible format, we infer them by aligning decoded token strings back to the original surface form. This is especially important for tokenizers with different conventions for whitespace markers, byte-level symbols, or special continuation markers. The evaluator maps byte-level offsets to Unicode character boundaries and records partial UTF-8 spans as diagnostics rather than silently assigning an incorrect boundary.

5.3 Reproducibility

All evaluations are run over fixed CSV datasets: the lemma-level SKMS-derived table, the large generated inflected-form table, and the 1,000-item open SlovakTokenBench benchmark. The scripts export both form-level metrics and aggregated tokenizer summaries, including results by case, suffix, declension pattern, and benchmark split. The main reported comparison uses the same metric implementation across tokenizers. The benchmark subset is intended to make the evaluation reusable without requiring the full internal generated dataset. Small numerical differences, especially between adjacent top-ranked tokenizers, should be interpreted cautiously because we do not report statistical significance tests in this version.

The released benchmark is accompanied by a README file describing the dataset columns, tokenizer registry, supported evaluation modes, and commands for running full or partial evaluations.

6 Large-Scale Lemma-Level Results

6.1 Tokenizer Summary

The lemma-level experiment evaluates tokenizers on 30,913 Slovak noun entries derived from SKMS-style morphemic segmentation. This setting measures how tokenizer boundaries align with morphemic boundaries in dictionary forms, rather than with generated inflectional suffixes.

Table 1 summarizes the tokenizer-level results. The best MBCR scores are achieved by `openai_cl100k_base` and `qwen3_8b`, although their difference is very small. In contrast, `skmt_morph_blm` has the lowest fertility, but not the best boundary preservation. One explanation is its Slovak-oriented vocabulary: SKMT can represent Slovak words using larger tokens, sometimes even as a single token.

Tokenizer	Tok. count	Fertility	MBCR
<code>openai_cl100k_base</code>	4.223	0.492	0.482
<code>qwen3_8b</code>	4.117	0.479	0.484
<code>llama_3_1_8b</code>	3.834	0.446	0.511
<code>mistral_nemo_tekken</code>	3.632	0.425	0.527
<code>openai_o200k_base</code>	3.525	0.413	0.543
<code>llama_4_maverick</code>	3.425	0.402	0.548
<code>gemma_3_270m</code>	3.399	0.398	0.549
<code>mdeberta_v3_base</code>	2.923	0.350	0.595
<code>xlm_roberta_base</code>	2.891	0.343	0.614
<code>mbert_cased</code>	3.425	0.404	0.651
<code>skmt_morph_blm</code>	2.092	0.255	0.662
<code>slovakbert</code>	3.140	0.371	0.682

Table 1: Lemma-level tokenizer summary. Lower fertility indicates more compact tokenization; lower MBCR indicates better morphemic boundary preservation.

6.2 Most Problematic Lemmas

The most problematic lemma-level examples tend to include spelling variants, slash variants, compounds, diacritics, and longer derived forms. These cases are useful for error inspection, but we treat them only as qualitative evidence because the inflected-form experiment provides the main stem-suffix evaluation.

6.3 Efficiency and Boundary Preservation

The lemma-level experiment supports the main claim of our paper: efficiency and morphological fidelity should be evaluated separately. Token count and fertility describe compactness, while MBCR describes whether tokenizer spans respect reference morphemic boundaries.

7 Large-Scale Inflected-Form Results

7.1 Overall Inflected-Form Results

The inflected-form experiment is the central large-scale evaluation. It uses 332,931 generated Slovak noun forms, covering 200,278 unique surface forms and 25,906 lemmas. Unlike the lemma-level experiment, this setting evaluates the practical stem-suffix boundary in declined forms.

The overall ranking is similar to the lemma-level results. `openai_cl100k_base` and `qwen3_8b` again have the lowest MBCR, while `skmt_morph_blm` is the most compact tokenizer but does not preserve stem-suffix boundaries best. This ranking should not be

Tokenizer	Tok. count	Fertility	Strict	MBCR
openai_cl100k_base	4.594	0.478	0.475	0.501
qwen3_8b	4.492	0.468	0.498	0.502
mistral_nemo_tekken	4.052	0.424	0.425	0.575
llama_3_1_8b	4.164	0.434	0.405	0.595
openai_o200k_base	3.937	0.411	0.385	0.615
llama_4_maverick	3.844	0.402	0.375	0.625
gemma_3_270m	3.818	0.399	0.360	0.640
mdeberta_v3_base	3.265	0.348	0.288	0.712
mbert_cased	3.810	0.400	0.285	0.715
xlm_roberta_base	3.279	0.346	0.273	0.727
skmt_morph_blm	2.475	0.267	0.236	0.764
slovakbert	3.427	0.361	0.102	0.898

Table 2: Large-scale inflected-form results. Strict is the strict stem-suffix boundary match. Lower MBCR is better; higher strict match is better.

read as direct evidence of stronger Slovak linguistic knowledge; higher boundary scores can also arise when a tokenizer splits words into more pieces.

An additional example of the efficiency-alignment trade-off appears in the two OpenAI tokenizers. The newer o200k_base tokenizer is more compact than cl100k_base, with lower fertility. However, cl100k_base achieves better MBCR. This suggests that vocabulary expansion or improved compression does not automatically lead to better morphological boundary preservation.

7.2 Results by Case

Case-level results show that not all inflectional forms are equally difficult under a boundary-preservation metric. Averaged across tokenizers, singular nominative has the highest MBCR (0.905), followed by singular genitive (0.840), plural nominative (0.809), singular accusative (0.801), singular locative (0.800). The easiest forms are plural instrumental (0.377), plural dative (0.408), plural locative (0.494), and singular instrumental (0.499).

The high value for singular nominative is not because these forms are hard to tokenize in general. Rather, nominative singular forms are often lemma-like and frequent, so tokenizers tend to encode them as whole words or large subword units. This is efficient, but it hides short stem-suffix boundaries such as *žen-a* or *mest-o*. Longer plural endings, by contrast, are more likely to become visible token material.

7.3 Results by Number

Plural forms are easier on average than singular forms. Across tokenizers, plural forms have an average MBCR of 0.569, while singular forms have an average MBCR of 0.768. This difference is largely explained by suffix length and distinctiveness: plural endings such as -ami, -och, or -om are more likely to be separated than short singular endings.

7.4 Results by Paradigm Pattern

The results also vary by declension pattern. Patterns with short endings, mobile vowels, or stem alternations tend to be harder. Among the high-MBCR patterns are types such

as *výsledok*, *kúsok*, *vajíčko*, *koniec*, and *chlapec*. Easier patterns tend to contain longer or more distinctive suffix material, where the stem–suffix boundary is more likely to coincide with a token boundary.

We treat pattern-level results as diagnostic rather than as a ranking of Slovak paradigms. Some patterns are represented by many more forms than others, and some generated forms are heuristic results requiring further validation.

7.5 Hard Inflected Forms

The hardest inflected forms often involve long derived words, compounds, diacritics, and short endings. Examples in the problematic lists include forms based on lemmas such as *ľahkomyselníčka*, *päťstokorunáčka*, *černokňazníctvo*, *krasokorčuľovanie*, and *zjednoslovnovanie*. These examples are difficult not only because they are long, but because their stems contain several internal subword-like regions before a short grammatical ending.

7.6 Lemma-Only vs. Inflected-Form Evaluation

Lemma-level and inflected-form evaluation answer different questions. The lemma-level experiment tests alignment with SKMS morphemic structure in dictionary forms. The inflected-form experiment tests whether tokenizers preserve the stem–suffix boundary that is relevant for Slovak declension.

Both experiments support the same general conclusion: compression and morphological alignment diverge. However, the inflected-form experiment is more directly connected to Slovak grammar, because case and number are expressed through suffixes in declined forms.

8 Benchmark Construction and Results

8.1 Design and Scope

To make the evaluation reproducible without requiring the full internal dataset, we construct SlovakTokenBench, a fixed 1,000-item benchmark of Slovak noun forms. Each item contains an inflected form together with grammatical metadata and a reference stem–suffix boundary. A separate set of 50 simple examples is included for quick tests and inspection and is not part of the scored total.

The original benchmark forms were manually checked to ensure that the included forms and paradigm assignments were suitable for evaluation. The corrected release keeps the manually checked surface form as authoritative and rebuilds the derived stem, suffix, and surface boundary fields from it. The additional challenge forms are selected from accepted paradigm entries and pass reconstruction checks.

The benchmark is intentionally not a purely random sample. It combines 300 random general examples, 270 paradigm-representative examples, 167 previously identified hard tokenization cases, 140 morphological edge cases, and 123 challenge replacement cases. These counts form the 1,000 scored items; the 50 simple examples are reported separately. They balance broad coverage with targeted stress cases while keeping the benchmark small enough for manual inspection.

Its purpose is to compare how tokenizers segment Slovak noun forms, especially whether they preserve stem–suffix boundaries. The benchmark supports reproducible

boundary-aware evaluation, while the larger generated dataset is used for broader aggregate trends.

8.2 Overall Results

Table 3 shows the overall benchmark results. The ranking remains consistent with the large-scale inflected-form experiment: `openai_cl100k_base` and `qwen3_8b` achieve the lowest MBCR, while `skmt_morph_blm` has the lowest fertility but weaker boundary preservation. The values for the corrected benchmark are computed from the 1,000 scored forms; zero-ending forms contribute to token-count and fertility means, while boundary-rate means use forms with an overt stem-suffix boundary.

Tokenizer	Tok. count	Fertility	Strict	MBCR
<code>openai_cl100k_base</code>	5.282	0.504	0.498	0.502
<code>qwen3_8b</code>	5.051	0.482	0.493	0.507
<code>mistral_nemo_tekken</code>	4.571	0.439	0.457	0.543
<code>llama_3_1_8b</code>	4.466	0.449	0.361	0.575
<code>openai_o200k_base</code>	4.450	0.426	0.426	0.574
<code>mbert_cased</code>	4.370	0.419	0.356	0.644
<code>llama_4_maverick</code>	4.064	0.411	0.314	0.621
<code>gemma_3_270m</code>	4.053	0.410	0.301	0.634
<code>mdeberta_v3_base</code>	3.667	0.358	0.268	0.732
<code>xlm_roberta_base</code>	3.724	0.360	0.268	0.732
<code>skmt_morph_blm</code>	2.444	0.258	0.245	0.691
<code>slovakbert</code>	3.679	0.357	0.126	0.874

Table 3: Overall SlovakTokenBench results. Strict is the strict stem-suffix boundary match. Lower MBCR is better; higher strict match is better. Values are means rounded to three decimals.

8.3 Interpretation

The benchmark confirms the main pattern from the large-scale experiments. The hard tokenization and challenge-replacement splits are especially challenging because they contain long derived forms, compounds, diacritics, stem alternations, and short endings.

Overall, SlovakTokenBench reproduces the central result of the full evaluation in a smaller and reusable form: efficiency and morphological alignment must be measured separately.

9 Discussion and Limitations

The main result of our evaluation is that fewer tokens do not necessarily mean better morphological tokenization. Across lemma-level, inflected-form, and benchmark experiments, compact tokenizers often achieve low fertility by using larger tokens, but these larger tokens may cross morphemic or stem-suffix boundaries. This is especially visible for `skmt_morph_blm`, which is consistently the most compact tokenizer but not the strongest boundary-preserving tokenizer under our metrics. The comparison between `openai_cl100k_base` and `openai_o200k_base` shows a similar pattern: the newer

o200k_base tokenizer is more compact, but cl100k_base preserves evaluated Slovak morphological boundaries better.

The strong MBCR performance of openai_cl100k_base and qwen3_8b should therefore be interpreted carefully. These tokenizers are not necessarily more linguistically aware of Slovak morphology. A likely explanation is an over-segmentation effect: tokenizers that split Slovak words into more pieces create more internal token boundaries, increasing the chance that one of them coincides with a morphemic or stem-suffix boundary. High boundary preservation can therefore result from fragmentation, while Slovak-oriented tokenizers may hide boundaries precisely because they encode larger Slovak units more compactly.

From a modeling perspective, boundary-aligned tokenization may make morphological regularities more explicit by allowing stems and suffixes to recur as shared units across related word forms. Compact tokenization, by contrast, may reduce sequence length while representing related forms as larger, less comparable units. This is a motivation for downstream testing rather than a conclusion about model accuracy.

The results also support the need for language-specific benchmarks. Slovak noun morphology is useful because grammatical information is often encoded in short suffixes, and these suffixes are easily absorbed into larger subword units. The same general issue is likely relevant for other Slavic languages with rich inflectional systems.

Several limitations should be noted. The large inflected-form dataset is generated heuristically and should not be treated as a fully manually validated morphological dictionary; we do not provide a global error-rate estimate for this generated set. The benchmark is a curated stress set rather than a population sample, and the accepted challenge replacements should be spot-checked against the current JÚLŠ noun database before a final linguistic release. SKMS morphemic segmentation and generated stem-suffix boundaries represent related but different linguistic layers, so the lemma-level and inflected-form experiments should not be collapsed into a single notion of morphology. Tokenizer offsets are not always exposed in the same way, and for some tokenizers they must be inferred by aligning decoded tokens back to the original form. Lastly, our evaluation is tokenizer-level only: it measures boundary preservation, not downstream model accuracy. Whether better boundary alignment improves Slovak generation, tagging, or grammatical reasoning remains a question for future work.

10 Conclusion and Future Work

This paper evaluated tokenization mismatch in Slovak noun morphology using lemma-level morphemic boundaries, large-scale generated inflected forms, and the manually checked SlovakTokenBench benchmark. The results show that tokenization efficiency and morphological boundary preservation are distinct dimensions of tokenizer quality across all three evaluation settings.

The practical conclusion is that tokenizer evaluation for morphologically rich languages should go beyond token count and fertility. Boundary-aware metrics such as MBCR, strict stem-suffix match, and boundary F1 provide a clearer view of whether tokenizers preserve linguistically meaningful structure.

Future work should test whether these boundary-alignment scores correlate with downstream Slovak NLP tasks, especially noun-form generation, morphological tagging, and constrained decoding. Another natural extension is to adapt the benchmark and evaluation pipeline to related Slavic languages. Such experiments would test whether the in-

trinsic over-segmentation and boundary-preservation effects observed here translate into measurable differences in model behavior.

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